

79 LL DMA Generic Driver

79.1 DMA Firmware driver registers structures

79.1.1 LL_DMA_InitTypeDef

LL_DMA_InitTypeDef is defined in the stm32f4xx_ll_dma.h

Data Fields

- **uint32_t PeriphOrM2MSrcAddress**
- **uint32_t MemoryOrM2MDstAddress**
- **uint32_t Direction**
- **uint32_t Mode**
- **uint32_t PeriphOrM2MSrcIncMode**
- **uint32_t MemoryOrM2MDstIncMode**
- **uint32_t PeriphOrM2MSrcDataSize**
- **uint32_t MemoryOrM2MDstDataSize**
- **uint32_t NbData**
- **uint32_t Channel**
- **uint32_t Priority**
- **uint32_t FIFOMode**
- **uint32_t FIFOThreshold**
- **uint32_t MemBurst**
- **uint32_t PeriphBurst**

Field Documentation

- **uint32_t LL_DMA_InitTypeDef::PeriphOrM2MSrcAddress**
Specifies the peripheral base address for DMA transfer or as Source base address in case of memory to memory transfer direction. This parameter must be a value between Min_Data = 0 and Max_Data = 0xFFFFFFFF.
- **uint32_t LL_DMA_InitTypeDef::MemoryOrM2MDstAddress**
Specifies the memory base address for DMA transfer or as Destination base address in case of memory to memory transfer direction. This parameter must be a value between Min_Data = 0 and Max_Data = 0xFFFFFFFF.
- **uint32_t LL_DMA_InitTypeDef::Direction**
Specifies if the data will be transferred from memory to peripheral, from memory to memory or from peripheral to memory. This parameter can be a value of [DMA_LL_EC_DIRECTION](#). This feature can be modified afterwards using unitary function [LL_DMA_SetDataTransferDirection\(\)](#).
- **uint32_t LL_DMA_InitTypeDef::Mode**
Specifies the normal or circular operation mode. This parameter can be a value of [DMA_LL_EC_MODE](#)
Note:
 - The circular buffer mode cannot be used if the memory to memory data transfer direction is configured on the selected Stream
This feature can be modified afterwards using unitary function [LL_DMA_SetMode\(\)](#).
- **uint32_t LL_DMA_InitTypeDef::PeriphOrM2MSrcIncMode**
Specifies whether the Peripheral address or Source address in case of memory to memory transfer direction is incremented or not. This parameter can be a value of [DMA_LL_EC_PERIPH](#). This feature can be modified afterwards using unitary function [LL_DMA_SetPeriphIncMode\(\)](#).
- **uint32_t LL_DMA_InitTypeDef::MemoryOrM2MDstIncMode**
Specifies whether the Memory address or Destination address in case of memory to memory transfer direction is incremented or not. This parameter can be a value of [DMA_LL_EC_MEMORY](#). This feature can be modified afterwards using unitary function [LL_DMA_SetMemoryIncMode\(\)](#).

- **`uint32_t LL_DMA_InitTypeDef::PeriphOrM2MSrcDataSize`**
Specifies the Peripheral data size alignment or Source data size alignment (byte, half word, word) in case of memory to memory transfer direction. This parameter can be a value of `DMA_LL_EC_PDATAALIGN`. This feature can be modified afterwards using unitary function `LL_DMA_SetPeriphSize()`.
- **`uint32_t LL_DMA_InitTypeDef::MemoryOrM2MDstDataSize`**
Specifies the Memory data size alignment or Destination data size alignment (byte, half word, word) in case of memory to memory transfer direction. This parameter can be a value of `DMA_LL_EC_MDATAALIGN`. This feature can be modified afterwards using unitary function `LL_DMA_SetMemorySize()`.
- **`uint32_t LL_DMA_InitTypeDef::NbData`**
Specifies the number of data to transfer, in data unit. The data unit is equal to the source buffer configuration set in `PeriphSize` or `MemorySize` parameters depending in the transfer direction. This parameter must be a value between `Min_Data = 0` and `Max_Data = 0x0000FFFF`. This feature can be modified afterwards using unitary function `LL_DMA_SetDataLength()`.
- **`uint32_t LL_DMA_InitTypeDef::Channel`**
Specifies the peripheral channel. This parameter can be a value of `DMA_LL_EC_CHANNEL`. This feature can be modified afterwards using unitary function `LL_DMA_SetChannelSelection()`.
- **`uint32_t LL_DMA_InitTypeDef::Priority`**
Specifies the channel priority level. This parameter can be a value of `DMA_LL_EC_PRIORITY`. This feature can be modified afterwards using unitary function `LL_DMA_SetStreamPriorityLevel()`.
- **`uint32_t LL_DMA_InitTypeDef::FIFOMode`**
Specifies if the FIFO mode or Direct mode will be used for the specified stream. This parameter can be a value of `DMA_LL_FIFOMODE`.
Note:
 - The Direct mode (FIFO mode disabled) cannot be used if the memory-to-memory data transfer is configured on the selected stream
This feature can be modified afterwards using unitary functions `LL_DMA_EnableFifoMode()` or `LL_DMA_EnableFifoMode()`.
- **`uint32_t LL_DMA_InitTypeDef::FIFOThreshold`**
Specifies the FIFO threshold level. This parameter can be a value of `DMA_LL_EC_FIFOTHRESHOLD`. This feature can be modified afterwards using unitary function `LL_DMA_SetFIFOThreshold()`.
- **`uint32_t LL_DMA_InitTypeDef::MemBurst`**
Specifies the Burst transfer configuration for the memory transfers. It specifies the amount of data to be transferred in a single non interruptible transaction. This parameter can be a value of `DMA_LL_EC_MBURST`.
Note:
 - The burst mode is possible only if the address Increment mode is enabled.
This feature can be modified afterwards using unitary function `LL_DMA_SetMemoryBurstxfer()`.
- **`uint32_t LL_DMA_InitTypeDef::PeriphBurst`**
Specifies the Burst transfer configuration for the peripheral transfers. It specifies the amount of data to be transferred in a single non interruptible transaction. This parameter can be a value of `DMA_LL_EC_PBURST`.
Note:
 - The burst mode is possible only if the address Increment mode is enabled.
This feature can be modified afterwards using unitary function `LL_DMA_SetPeriphBurstxfer()`.

79.2 DMA Firmware driver API description

The following section lists the various functions of the DMA library.

79.2.1 Detailed description of functions

LL_DMA_EnableStream

Function name

```
__STATIC_INLINE void LL_DMA_EnableStream (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Enable DMA stream.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR EN LL_DMA_EnableStream

LL_DMA_DisableStream

Function name

```
__STATIC_INLINE void LL_DMA_DisableStream (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Disable DMA stream.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR EN LL_DMA_DisableStream

LL_DMA_IsEnabledStream

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsEnabledStream (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Check if DMA stream is enabled or disabled.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- CR EN LL_DMA_IsEnabledStream

LL_DMA_ConfigTransfer

Function name

__STATIC_INLINE void LL_DMA_ConfigTransfer (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t Configuration)

Function description

Configure all parameters linked to DMA transfer.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **Configuration:** This parameter must be a combination of all the following values:
 - LL_DMA_DIRECTION_PERIPH_TO_MEMORY or LL_DMA_DIRECTION_MEMORY_TO_PERIPH or LL_DMA_DIRECTION_MEMORY_TO_MEMORY
 - LL_DMA_MODE_NORMAL or LL_DMA_MODE_CIRCULAR or LL_DMA_MODE_PFCTRL
 - LL_DMA_PERIPH_INCREMENT or LL_DMA_PERIPH_NOINCREMENT
 - LL_DMA_MEMORY_INCREMENT or LL_DMA_MEMORY_NOINCREMENT
 - LL_DMA_PDATAALIGN_BYTE or LL_DMA_PDATAALIGN_HALFWORD or LL_DMA_PDATAALIGN_WORD
 - LL_DMA_MDATAALIGN_BYTE or LL_DMA_MDATAALIGN_HALFWORD or LL_DMA_MDATAALIGN_WORD
 - LL_DMA_PRIORITY_LOW or LL_DMA_PRIORITY_MEDIUM or LL_DMA_PRIORITY_HIGH or LL_DMA_PRIORITY_VERYHIGH

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR DIR LL_DMA_ConfigTransfer
- CR CIRC LL_DMA_ConfigTransfer
- CR PINC LL_DMA_ConfigTransfer
- CR MINC LL_DMA_ConfigTransfer
- CR PSIZE LL_DMA_ConfigTransfer
- CR MSIZE LL_DMA_ConfigTransfer
- CR PL LL_DMA_ConfigTransfer
- CR PFCTRL LL_DMA_ConfigTransfer

LL_DMA_SetDataTransferDirection

Function name

```
__STATIC_INLINE void LL_DMA_SetDataTransferDirection (DMA_TypeDef * DMAx, uint32_t Stream,
uint32_t Direction)
```

Function description

Set Data transfer direction (read from peripheral or from memory).

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **Direction:** This parameter can be one of the following values:
 - LL_DMA_DIRECTION_PERIPH_TO_MEMORY
 - LL_DMA_DIRECTION_MEMORY_TO_PERIPH
 - LL_DMA_DIRECTION_MEMORY_TO_MEMORY

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR DIR LL_DMA_SetDataTransferDirection

LL_DMA_GetDataTransferDirection

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetDataTransferDirection (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get Data transfer direction (read from peripheral or from memory).

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_DIRECTION_PERIPH_TO_MEMORY
 - LL_DMA_DIRECTION_MEMORY_TO_PERIPH
 - LL_DMA_DIRECTION_MEMORY_TO_MEMORY

Reference Manual to LL API cross reference:

- CR DIR LL_DMA_GetDataTransferDirection

LL_DMA_SetMode

Function name

__STATIC_INLINE void LL_DMA_SetMode (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t Mode)

Function description

Set DMA mode normal, circular or peripheral flow control.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **Mode:** This parameter can be one of the following values:
 - LL_DMA_MODE_NORMAL
 - LL_DMA_MODE_CIRCULAR
 - LL_DMA_MODE_PFCTRL

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR CIRC LL_DMA_SetMode
- CR PFCTRL LL_DMA_SetMode

LL_DMA_GetMode

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetMode (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get DMA mode normal, circular or peripheral flow control.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_MODE_NORMAL
 - LL_DMA_MODE_CIRCULAR
 - LL_DMA_MODE_PFCTRL

Reference Manual to LL API cross reference:

- CR CIRC LL_DMA_GetMode
- CR PFCTRL LL_DMA_GetMode

LL_DMA_SetPeriphIncMode

Function name

```
__STATIC_INLINE void LL_DMA_SetPeriphIncMode (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t IncrementMode)
```

Function description

Set Peripheral increment mode.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **IncrementMode:** This parameter can be one of the following values:
 - LL_DMA_PERIPH_NOINCREMENT
 - LL_DMA_PERIPH_INCREMENT

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR PINC LL_DMA_SetPeriphIncMode

LL_DMA_GetPeriphIncMode

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetPeriphIncMode (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get Peripheral increment mode.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_PERIPH_NOINCREMENT
 - LL_DMA_PERIPH_INCREMENT

Reference Manual to LL API cross reference:

- CR PINC LL_DMA_GetPeriphIncMode

LL_DMA_SetMemoryIncMode

Function name

```
__STATIC_INLINE void LL_DMA_SetMemoryIncMode (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t IncrementMode)
```

Function description

Set Memory increment mode.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **IncrementMode:** This parameter can be one of the following values:
 - LL_DMA_MEMORY_NOINCREMENT
 - LL_DMA_MEMORY_INCREMENT

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR MINC LL_DMA_SetMemoryIncMode

LL_DMA_GetMemoryIncMode

Function name

__STATIC_INLINE uint32_t LL_DMA_GetMemoryIncMode (DMA_TypeDef * DMAx, uint32_t Stream)

Function description

Get Memory increment mode.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_MEMORY_NOINCREMENT
 - LL_DMA_MEMORY_INCREMENT

Reference Manual to LL API cross reference:

- CR MINC LL_DMA_GetMemoryIncMode

LL_DMA_SetPeriphSize

Function name

__STATIC_INLINE void LL_DMA_SetPeriphSize (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t Size)

Function description

Set Peripheral size.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **Size:** This parameter can be one of the following values:
 - LL_DMA_PDATAALIGN_BYTE
 - LL_DMA_PDATAALIGN_HALFWORD
 - LL_DMA_PDATAALIGN_WORD

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR PSIZE LL_DMA_SetPeriphSize

LL_DMA_GetPeriphSize

Function name

`__STATIC_INLINE uint32_t LL_DMA_GetPeriphSize (DMA_TypeDef * DMAx, uint32_t Stream)`

Function description

Get Peripheral size.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_PDATAALIGN_BYTE
 - LL_DMA_PDATAALIGN_HALFWORD
 - LL_DMA_PDATAALIGN_WORD

Reference Manual to LL API cross reference:

- CR PSIZE LL_DMA_GetPeriphSize

LL_DMA_SetMemorySize

Function name

```
__STATIC_INLINE void LL_DMA_SetMemorySize (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t Size)
```

Function description

Set Memory size.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **Size:** This parameter can be one of the following values:
 - LL_DMA_MDATAALIGN_BYTE
 - LL_DMA_MDATAALIGN_HALFWORD
 - LL_DMA_MDATAALIGN_WORD

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR MSIZE LL_DMA_SetMemorySize

LL_DMA_GetMemorySize

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetMemorySize (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get Memory size.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_MDATAALIGN_BYTE
 - LL_DMA_MDATAALIGN_HALFWORD
 - LL_DMA_MDATAALIGN_WORD

Reference Manual to LL API cross reference:

- CR MSIZE LL_DMA_GetMemorySize

LL_DMA_SetIncOffsetSize

Function name

__STATIC_INLINE void LL_DMA_SetIncOffsetSize (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t OffsetSize)

Function description

Set Peripheral increment offset size.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **OffsetSize:** This parameter can be one of the following values:
 - LL_DMA_OFFSETSIZE_PSIZE
 - LL_DMA_OFFSETSIZE_FIXEDTO4

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR PINCOS LL_DMA_SetIncOffsetSize

LL_DMA_GetIncOffsetSize

Function name

__STATIC_INLINE uint32_t LL_DMA_GetIncOffsetSize (DMA_TypeDef * DMAx, uint32_t Stream)

Function description

Get Peripheral increment offset size.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_OFFSETSIZE_PSIZE
 - LL_DMA_OFFSETSIZE_FIXEDTO4

Reference Manual to LL API cross reference:

- CR PINCOS LL_DMA_GetIncOffsetSize

LL_DMA_SetStreamPriorityLevel

Function name

```
__STATIC_INLINE void LL_DMA_SetStreamPriorityLevel (DMA_TypeDef * DMAx, uint32_t Stream,
uint32_t Priority)
```

Function description

Set Stream priority level.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **Priority:** This parameter can be one of the following values:
 - LL_DMA_PRIORITY_LOW
 - LL_DMA_PRIORITY_MEDIUM
 - LL_DMA_PRIORITY_HIGH
 - LL_DMA_PRIORITY_VERYHIGH

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR PL LL_DMA_SetStreamPriorityLevel

LL_DMA_GetStreamPriorityLevel

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetStreamPriorityLevel (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get Stream priority level.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_PRIORITY_LOW
 - LL_DMA_PRIORITY_MEDIUM
 - LL_DMA_PRIORITY_HIGH
 - LL_DMA_PRIORITY_VERYHIGH

Reference Manual to LL API cross reference:

- CR PL LL_DMA_GetStreamPriorityLevel

LL_DMA_SetDataLength

Function name

```
__STATIC_INLINE void LL_DMA_SetDataLength (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t NbData)
```

Function description

Set Number of data to transfer.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **NbData:** Between 0 to 0xFFFFFFFF

Return values

- **None:**

Notes

- This action has no effect if stream is enabled.

Reference Manual to LL API cross reference:

- NDTR NDT LL_DMA_SetDataLength

LL_DMA_GetDataLength

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetDataLength (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get Number of data to transfer.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Between:** 0 to 0xFFFFFFFF

Notes

- Once the stream is enabled, the return value indicate the remaining bytes to be transmitted.

Reference Manual to LL API cross reference:

- NDTR NDT LL_DMA_GetDataLength

LL_DMA_SetChannelSelection

Function name

```
__STATIC_INLINE void LL_DMA_SetChannelSelection (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t Channel)
```

Function description

Select Channel number associated to the Stream.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **Channel:** This parameter can be one of the following values:
 - LL_DMA_CHANNEL_0
 - LL_DMA_CHANNEL_1
 - LL_DMA_CHANNEL_2
 - LL_DMA_CHANNEL_3
 - LL_DMA_CHANNEL_4
 - LL_DMA_CHANNEL_5
 - LL_DMA_CHANNEL_6
 - LL_DMA_CHANNEL_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR CHSEL LL_DMA_SetChannelSelection

LL_DMA_GetChannelSelection

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetChannelSelection (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get the Channel number associated to the Stream.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_CHANNEL_0
 - LL_DMA_CHANNEL_1
 - LL_DMA_CHANNEL_2
 - LL_DMA_CHANNEL_3
 - LL_DMA_CHANNEL_4
 - LL_DMA_CHANNEL_5
 - LL_DMA_CHANNEL_6
 - LL_DMA_CHANNEL_7

Reference Manual to LL API cross reference:

- CR CHSEL LL_DMA_GetChannelSelection

LL_DMA_SetMemoryBurstxfer

Function name

```
__STATIC_INLINE void LL_DMA_SetMemoryBurstxfer (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t Mburst)
```

Function description

Set Memory burst transfer configuration.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **Mburst:** This parameter can be one of the following values:
 - LL_DMA_MBURST_SINGLE
 - LL_DMA_MBURST_INC4
 - LL_DMA_MBURST_INC8
 - LL_DMA_MBURST_INC16

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR MBURST LL_DMA_SetMemoryBurstxfer

LL_DMA_GetMemoryBurstxfer

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetMemoryBurstxfer (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get Memory burst transfer configuration.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_MBURST_SINGLE
 - LL_DMA_MBURST_INC4
 - LL_DMA_MBURST_INC8
 - LL_DMA_MBURST_INC16

Reference Manual to LL API cross reference:

- CR MBURST LL_DMA_GetMemoryBurstxfer

LL_DMA_SetPeriphBurstxfer

Function name

```
__STATIC_INLINE void LL_DMA_SetPeriphBurstxfer (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t Pburst)
```

Function description

Set Peripheral burst transfer configuration.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **Pburst:** This parameter can be one of the following values:
 - LL_DMA_PBURST_SINGLE
 - LL_DMA_PBURST_INC4
 - LL_DMA_PBURST_INC8
 - LL_DMA_PBURST_INC16

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR PBURST LL_DMA_SetPeriphBurstxfer

LL_DMA_GetPeriphBurstxfer

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetPeriphBurstxfer (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get Peripheral burst transfer configuration.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_PBURST_SINGLE
 - LL_DMA_PBURST_INC4
 - LL_DMA_PBURST_INC8
 - LL_DMA_PBURST_INC16

Reference Manual to LL API cross reference:

- CR PBURST LL_DMA_GetPeriphBurstxfer

LL_DMA_SetCurrentTargetMem

Function name

```
__STATIC_INLINE void LL_DMA_SetCurrentTargetMem (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t CurrentMemory)
```

Function description

Set Current target (only in double buffer mode) to Memory 1 or Memory 0.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **CurrentMemory:** This parameter can be one of the following values:
 - LL_DMA_CURRENTTARGETMEM0
 - LL_DMA_CURRENTTARGETMEM1

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR CT LL_DMA_SetCurrentTargetMem

LL_DMA_GetCurrentTargetMem

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetCurrentTargetMem (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Set Current target (only in double buffer mode) to Memory 1 or Memory 0.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_CURRENTTARGETMEM0
 - LL_DMA_CURRENTTARGETMEM1

Reference Manual to LL API cross reference:

- CR CT LL_DMA_GetCurrentTargetMem

LL_DMA_EnableDoubleBufferMode

Function name

```
__STATIC_INLINE void LL_DMA_EnableDoubleBufferMode (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Enable the double buffer mode.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR DBM LL_DMA_EnableDoubleBufferMode

LL_DMA_DisableDoubleBufferMode

Function name

```
__STATIC_INLINE void LL_DMA_DisableDoubleBufferMode (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Disable the double buffer mode.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR DBM LL_DMA_DisableDoubleBufferMode

LL_DMA_GetFIFOStatus

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetFIFOStatus (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get FIFO status.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_FIFOSTATUS_0_25
 - LL_DMA_FIFOSTATUS_25_50
 - LL_DMA_FIFOSTATUS_50_75
 - LL_DMA_FIFOSTATUS_75_100
 - LL_DMA_FIFOSTATUS_EMPTY
 - LL_DMA_FIFOSTATUS_FULL

Reference Manual to LL API cross reference:

- FCR FS LL_DMA_GetFIFOStatus

LL_DMA_DisableFifoMode

Function name

```
__STATIC_INLINE void LL_DMA_DisableFifoMode (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Disable Fifo mode.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- FCR DMDIS LL_DMA_DisableFifoMode

LL_DMA_EnableFifoMode

Function name

```
__STATIC_INLINE void LL_DMA_EnableFifoMode (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Enable Fifo mode.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- FCR DMDIS LL_DMA_EnableFifoMode

LL_DMA_SetFIFOThreshold

Function name

```
__STATIC_INLINE void LL_DMA_SetFIFOThreshold (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t Threshold)
```

Function description

Select FIFO threshold.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **Threshold:** This parameter can be one of the following values:
 - LL_DMA_FIFOTHRESHOLD_1_4
 - LL_DMA_FIFOTHRESHOLD_1_2
 - LL_DMA_FIFOTHRESHOLD_3_4
 - LL_DMA_FIFOTHRESHOLD_FULL

Return values

- **None:**

Reference Manual to LL API cross reference:

- FCR FTH LL_DMA_SetFIFOThreshold

LL_DMA_GetFIFOThreshold

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetFIFOThreshold (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get FIFO threshold.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Returned:** value can be one of the following values:
 - LL_DMA_FIFOTHRESHOLD_1_4
 - LL_DMA_FIFOTHRESHOLD_1_2
 - LL_DMA_FIFOTHRESHOLD_3_4
 - LL_DMA_FIFOTHRESHOLD_FULL

Reference Manual to LL API cross reference:

- FCR FTH LL_DMA_GetFIFOThreshold

LL_DMA_ConfigFifo

Function name

```
__STATIC_INLINE void LL_DMA_ConfigFifo (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t FifoMode,
uint32_t FifoThreshold)
```

Function description

Configure the FIFO .

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **FifoMode:** This parameter can be one of the following values:
 - LL_DMA_FIFOMODE_ENABLE
 - LL_DMA_FIFOMODE_DISABLE
- **FifoThreshold:** This parameter can be one of the following values:
 - LL_DMA_FIFOTHRESHOLD_1_4
 - LL_DMA_FIFOTHRESHOLD_1_2
 - LL_DMA_FIFOTHRESHOLD_3_4
 - LL_DMA_FIFOTHRESHOLD_FULL

Return values

- **None:**

Reference Manual to LL API cross reference:

- FCR FTH LL_DMA_ConfigFifo
- FCR DMDIS LL_DMA_ConfigFifo

LL_DMA_ConfigAddresses

Function name

```
__STATIC_INLINE void LL_DMA_ConfigAddresses (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t SrcAddress, uint32_t DstAddress, uint32_t Direction)
```

Function description

Configure the Source and Destination addresses.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **SrcAddress:** Between 0 to 0xFFFFFFFF
- **DstAddress:** Between 0 to 0xFFFFFFFF
- **Direction:** This parameter can be one of the following values:
 - LL_DMA_DIRECTION_PERIPH_TO_MEMORY
 - LL_DMA_DIRECTION_MEMORY_TO_PERIPH
 - LL_DMA_DIRECTION_MEMORY_TO_MEMORY

Return values

- **None:**

Notes

- This API must not be called when the DMA stream is enabled.

Reference Manual to LL API cross reference:

- M0AR M0A LL_DMA_ConfigAddresses
- PAR PA LL_DMA_ConfigAddresses

LL_DMA_SetMemoryAddress

Function name

```
__STATIC_INLINE void LL_DMA_SetMemoryAddress (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t MemoryAddress)
```

Function description

Set the Memory address.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **MemoryAddress:** Between 0 to 0xFFFFFFFF

Return values

- **None:**

Notes

- Interface used for direction LL_DMA_DIRECTION_PERIPH_TO_MEMORY or LL_DMA_DIRECTION_MEMORY_TO_PERIPH only.
- This API must not be called when the DMA channel is enabled.

Reference Manual to LL API cross reference:

- M0AR M0A LL_DMA_SetMemoryAddress

LL_DMA_SetPeriphAddress

Function name

```
__STATIC_INLINE void LL_DMA_SetPeriphAddress (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t PeriphAddress)
```

Function description

Set the Peripheral address.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **PeriphAddress:** Between 0 to 0xFFFFFFFF

Return values

- **None:**

Notes

- Interface used for direction LL_DMA_DIRECTION_PERIPH_TO_MEMORY or LL_DMA_DIRECTION_MEMORY_TO_PERIPH only.
- This API must not be called when the DMA channel is enabled.

Reference Manual to LL API cross reference:

- PAR PA LL_DMA_SetPeriphAddress

LL_DMA_GetMemoryAddress

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetMemoryAddress (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get the Memory address.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Between:** 0 to 0xFFFFFFFF

Notes

- Interface used for direction LL_DMA_DIRECTION_PERIPH_TO_MEMORY or LL_DMA_DIRECTION_MEMORY_TO_PERIPH only.

Reference Manual to LL API cross reference:

- M0AR M0A LL_DMA_GetMemoryAddress

LL_DMA_GetPeriphAddress

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetPeriphAddress (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get the Peripheral address.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Between:** 0 to 0xFFFFFFFF

Notes

- Interface used for direction LL_DMA_DIRECTION_PERIPH_TO_MEMORY or LL_DMA_DIRECTION_MEMORY_TO_PERIPH only.

Reference Manual to LL API cross reference:

- PAR PA LL_DMA_GetPeriphAddress

LL_DMA_SetM2MSrcAddress

Function name

```
__STATIC_INLINE void LL_DMA_SetM2MSrcAddress (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t MemoryAddress)
```

Function description

Set the Memory to Memory Source address.

Parameters

- DMAx**: DMAx Instance
- Stream**: This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- MemoryAddress**: Between 0 to 0xFFFFFFFF

Return values

- None**:

Notes

- Interface used for direction LL_DMA_DIRECTION_MEMORY_TO_MEMORY only.
- This API must not be called when the DMA channel is enabled.

Reference Manual to LL API cross reference:

- PAR PA LL_DMA_SetM2MSrcAddress

LL_DMA_SetM2MDstAddress

Function name

```
__STATIC_INLINE void LL_DMA_SetM2MDstAddress (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t MemoryAddress)
```

Function description

Set the Memory to Memory Destination address.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **MemoryAddress:** Between 0 to 0xFFFFFFFF

Return values

- **None:**

Notes

- Interface used for direction LL_DMA_DIRECTION_MEMORY_TO_MEMORY only.
- This API must not be called when the DMA channel is enabled.

Reference Manual to LL API cross reference:

- M0AR M0A LL_DMA_SetM2MDstAddress

LL_DMA_GetM2MSrcAddress

Function name

__STATIC_INLINE uint32_t LL_DMA_GetM2MSrcAddress (DMA_TypeDef * DMAx, uint32_t Stream)

Function description

Get the Memory to Memory Source address.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Between:** 0 to 0xFFFFFFFF

Notes

- Interface used for direction LL_DMA_DIRECTION_MEMORY_TO_MEMORY only.

Reference Manual to LL API cross reference:

- PAR PA LL_DMA_GetM2MSrcAddress

LL_DMA_GetM2MDstAddress

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetM2MDstAddress (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get the Memory to Memory Destination address.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Between:** 0 to 0xFFFFFFFF

Notes

- Interface used for direction LL_DMA_DIRECTION_MEMORY_TO_MEMORY only.

Reference Manual to LL API cross reference:

- M0AR M0A LL_DMA_GetM2MDstAddress

LL_DMA_SetMemory1Address

Function name

```
__STATIC_INLINE void LL_DMA_SetMemory1Address (DMA_TypeDef * DMAx, uint32_t Stream, uint32_t Address)
```

Function description

Set Memory 1 address (used in case of Double buffer mode).

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **Address:** Between 0 to 0xFFFFFFFF

Return values

- **None:**

Reference Manual to LL API cross reference:

- M1AR M1A LL_DMA_SetMemory1Address

LL_DMA_GetMemory1Address

Function name

```
__STATIC_INLINE uint32_t LL_DMA_GetMemory1Address (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Get Memory 1 address (used in case of Double buffer mode).

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **Between:** 0 to 0xFFFFFFFF

Reference Manual to LL API cross reference:

- M1AR M1A LL_DMA_GetMemory1Address

LL_DMA_IsActiveFlag_HT0

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_HT0 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 0 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR HTIF0 LL_DMA_IsActiveFlag_HT0

LL_DMA_IsActiveFlag_HT1

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_HT1 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 1 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR HTIF1 LL_DMA_IsActiveFlag_HT1

LL_DMA_IsActiveFlag_HT2

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_HT2 (DMA_TypeDef * DMAx)

Function description

Get Stream 2 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR HTIF2 LL_DMA_IsActiveFlag_HT2

LL_DMA_IsActiveFlag_HT3

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_HT3 (DMA_TypeDef * DMAx)

Function description

Get Stream 3 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR HTIF3 LL_DMA_IsActiveFlag_HT3

LL_DMA_IsActiveFlag_HT4

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_HT4 (DMA_TypeDef * DMAx)

Function description

Get Stream 4 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR HTIF4 LL_DMA_IsActiveFlag_HT4

LL_DMA_IsActiveFlag_HT5

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_HT5 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 5 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR HTIF0 LL_DMA_IsActiveFlag_HT5

LL_DMA_IsActiveFlag_HT6

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_HT6 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 6 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR HTIF6 LL_DMA_IsActiveFlag_HT6

LL_DMA_IsActiveFlag_HT7

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_HT7 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 7 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR HTIF7 LL_DMA_IsActiveFlag_HT7

LL_DMA_IsActiveFlag_TC0

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TC0 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 0 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR TCIF0 LL_DMA_IsActiveFlag_TC0

LL_DMA_IsActiveFlag_TC1

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TC1 (DMA_TypeDef * DMAx)

Function description

Get Stream 1 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR TCIF1 LL_DMA_IsActiveFlag_TC1

LL_DMA_IsActiveFlag_TC2

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TC2 (DMA_TypeDef * DMAx)

Function description

Get Stream 2 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR TCIF2 LL_DMA_IsActiveFlag_TC2

LL_DMA_IsActiveFlag_TC3

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TC3 (DMA_TypeDef * DMAx)

Function description

Get Stream 3 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR TCIF3 LL_DMA_IsActiveFlag_TC3

LL_DMA_IsActiveFlag_TC4

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TC4 (DMA_TypeDef * DMAx)

Function description

Get Stream 4 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR TCIF4 LL_DMA_IsActiveFlag_TC4

LL_DMA_IsActiveFlag_TC5

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TC5 (DMA_TypeDef * DMAx)

Function description

Get Stream 5 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR TCIF0 LL_DMA_IsActiveFlag_TC5

LL_DMA_IsActiveFlag_TC6

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TC6 (DMA_TypeDef * DMAx)

Function description

Get Stream 6 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR TCIF6 LL_DMA_IsActiveFlag_TC6

LL_DMA_IsActiveFlag_TC7

Function name

`__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TC7 (DMA_TypeDef * DMAx)`

Function description

Get Stream 7 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR TCIF7 LL_DMA_IsActiveFlag_TC7

LL_DMA_IsActiveFlag_TE0

Function name

`__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TE0 (DMA_TypeDef * DMAx)`

Function description

Get Stream 0 transfer error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR TEIF0 LL_DMA_IsActiveFlag_TE0

LL_DMA_IsActiveFlag_TE1

Function name

`__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TE1 (DMA_TypeDef * DMAx)`

Function description

Get Stream 1 transfer error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR TEIF1 LL_DMA_IsActiveFlag_TE1

LL_DMA_IsActiveFlag_TE2

Function name

`__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TE2 (DMA_TypeDef * DMAx)`

Function description

Get Stream 2 transfer error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR TEIF2 LL_DMA_IsActiveFlag_TE2

LL_DMA_IsActiveFlag_TE3

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TE3 (DMA_TypeDef * DMAx)

Function description

Get Stream 3 transfer error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR TEIF3 LL_DMA_IsActiveFlag_TE3

LL_DMA_IsActiveFlag_TE4

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TE4 (DMA_TypeDef * DMAx)

Function description

Get Stream 4 transfer error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR TEIF4 LL_DMA_IsActiveFlag_TE4

LL_DMA_IsActiveFlag_TE5

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TE5 (DMA_TypeDef * DMAx)

Function description

Get Stream 5 transfer error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR TEIF0 LL_DMA_IsActiveFlag_TE5

LL_DMA_IsActiveFlag_TE6

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TE6 (DMA_TypeDef * DMAx)

Function description

Get Stream 6 transfer error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR TEIF6 LL_DMA_IsActiveFlag_TE6

LL_DMA_IsActiveFlag_TE7

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_TE7 (DMA_TypeDef * DMAx)

Function description

Get Stream 7 transfer error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR TEIF7 LL_DMA_IsActiveFlag_TE7

LL_DMA_IsActiveFlag_DME0

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_DME0 (DMA_TypeDef * DMAx)

Function description

Get Stream 0 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR DMEIF0 LL_DMA_IsActiveFlag_DME0

LL_DMA_IsActiveFlag_DME1

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_DME1 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 1 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR DMEIF1 LL_DMA_IsActiveFlag_DME1

LL_DMA_IsActiveFlag_DME2

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_DME2 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 2 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR DMEIF2 LL_DMA_IsActiveFlag_DME2

LL_DMA_IsActiveFlag_DME3

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_DME3 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 3 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR DMEIF3 LL_DMA_IsActiveFlag_DME3

LL_DMA_IsActiveFlag_DME4

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_DME4 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 4 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR DMEIF4 LL_DMA_IsActiveFlag_DME4

LL_DMA_IsActiveFlag_DME5

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_DME5 (DMA_TypeDef * DMAx)

Function description

Get Stream 5 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR DMEIF0 LL_DMA_IsActiveFlag_DME5

LL_DMA_IsActiveFlag_DME6

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_DME6 (DMA_TypeDef * DMAx)

Function description

Get Stream 6 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR DMEIF6 LL_DMA_IsActiveFlag_DME6

LL_DMA_IsActiveFlag_DME7

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_DME7 (DMA_TypeDef * DMAx)

Function description

Get Stream 7 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR DMEIF7 LL_DMA_IsActiveFlag_DME7

LL_DMA_IsActiveFlag_FE0

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_FE0 (DMA_TypeDef * DMAx)

Function description

Get Stream 0 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR FEIF0 LL_DMA_IsActiveFlag_FE0

LL_DMA_IsActiveFlag_FE1

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_FE1 (DMA_TypeDef * DMAx)

Function description

Get Stream 1 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR FEIF1 LL_DMA_IsActiveFlag_FE1

LL_DMA_IsActiveFlag_FE2

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_FE2 (DMA_TypeDef * DMAx)

Function description

Get Stream 2 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR FEIF2 LL_DMA_IsActiveFlag_FE2

LL_DMA_IsActiveFlag_FE3

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_FE3 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 3 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LISR FEIF3 LL_DMA_IsActiveFlag_FE3

LL_DMA_IsActiveFlag_FE4

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_FE4 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 4 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR FEIF4 LL_DMA_IsActiveFlag_FE4

LL_DMA_IsActiveFlag_FE5

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_FE5 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 5 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR FEIF0 LL_DMA_IsActiveFlag_FE5

LL_DMA_IsActiveFlag_FE6

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_FE6 (DMA_TypeDef * DMAx)
```

Function description

Get Stream 6 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR FEIF6 LL_DMA_IsActiveFlag_FE6

LL_DMA_IsActiveFlag_FE7

Function name

__STATIC_INLINE uint32_t LL_DMA_IsActiveFlag_FE7 (DMA_TypeDef * DMAx)

Function description

Get Stream 7 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- HISR FEIF7 LL_DMA_IsActiveFlag_FE7

LL_DMA_ClearFlag_HT0

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_HT0 (DMA_TypeDef * DMAx)

Function description

Clear Stream 0 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CHTIF0 LL_DMA_ClearFlag_HT0

LL_DMA_ClearFlag_HT1

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_HT1 (DMA_TypeDef * DMAx)

Function description

Clear Stream 1 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CHTIF1 LL_DMA_ClearFlag_HT1

LL_DMA_ClearFlag_HT2

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_HT2 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 2 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CHTIF2 LL_DMA_ClearFlag_HT2

LL_DMA_ClearFlag_HT3

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_HT3 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 3 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CHTIF3 LL_DMA_ClearFlag_HT3

LL_DMA_ClearFlag_HT4

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_HT4 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 4 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CHTIF4 LL_DMA_ClearFlag_HT4

LL_DMA_ClearFlag_HT5**Function name**

```
__STATIC_INLINE void LL_DMA_ClearFlag_HT5 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 5 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CHTIF5 LL_DMA_ClearFlag_HT5

LL_DMA_ClearFlag_HT6**Function name**

```
__STATIC_INLINE void LL_DMA_ClearFlag_HT6 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 6 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CHTIF6 LL_DMA_ClearFlag_HT6

LL_DMA_ClearFlag_HT7**Function name**

```
__STATIC_INLINE void LL_DMA_ClearFlag_HT7 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 7 half transfer flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CHTIF7 LL_DMA_ClearFlag_HT7

LL_DMA_ClearFlag_TC0**Function name**

```
__STATIC_INLINE void LL_DMA_ClearFlag_TC0 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 0 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CTCIF0 LL_DMA_ClearFlag_TC0

LL_DMA_ClearFlag_TC1

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_TC1 (DMA_TypeDef * DMAx)

Function description

Clear Stream 1 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CTCIF1 LL_DMA_ClearFlag_TC1

LL_DMA_ClearFlag_TC2

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_TC2 (DMA_TypeDef * DMAx)

Function description

Clear Stream 2 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CTCIF2 LL_DMA_ClearFlag_TC2

LL_DMA_ClearFlag_TC3

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_TC3 (DMA_TypeDef * DMAx)

Function description

Clear Stream 3 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CTCIF3 LL_DMA_ClearFlag_TC3

LL_DMA_ClearFlag_TC4

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_TC4 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 4 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CTCIF4 LL_DMA_ClearFlag_TC4

LL_DMA_ClearFlag_TC5

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_TC5 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 5 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CTCIF5 LL_DMA_ClearFlag_TC5

LL_DMA_ClearFlag_TC6

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_TC6 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 6 transfer complete flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CTCIF6 LL_DMA_ClearFlag_TC6

LL_DMA_ClearFlag_TC7

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_TC7 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 7 transfer complete flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CTCIF7 LL_DMA_ClearFlag_TC7

LL_DMA_ClearFlag_TE0

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_TE0 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 0 transfer error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CTEIF0 LL_DMA_ClearFlag_TE0

LL_DMA_ClearFlag_TE1

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_TE1 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 1 transfer error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CTEIF1 LL_DMA_ClearFlag_TE1

LL_DMA_ClearFlag_TE2

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_TE2 (DMA_TypeDef * DMAx)
```


Function description

Clear Stream 2 transfer error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None**:

Reference Manual to LL API cross reference:

- LIFCR CTEIF2 LL_DMA_ClearFlag_TE2

LL_DMA_ClearFlag_TE3

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_TE3 (DMA_TypeDef * DMAx)

Function description

Clear Stream 3 transfer error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None**:

Reference Manual to LL API cross reference:

- LIFCR CTEIF3 LL_DMA_ClearFlag_TE3

LL_DMA_ClearFlag_TE4

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_TE4 (DMA_TypeDef * DMAx)

Function description

Clear Stream 4 transfer error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None**:

Reference Manual to LL API cross reference:

- HIFCR CTEIF4 LL_DMA_ClearFlag_TE4

LL_DMA_ClearFlag_TE5

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_TE5 (DMA_TypeDef * DMAx)

Function description

Clear Stream 5 transfer error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CTEIF5 LL_DMA_ClearFlag_TE5

LL_DMA_ClearFlag_TE6

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_TE6 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 6 transfer error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CTEIF6 LL_DMA_ClearFlag_TE6

LL_DMA_ClearFlag_TE7

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_TE7 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 7 transfer error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CTEIF7 LL_DMA_ClearFlag_TE7

LL_DMA_ClearFlag_DME0

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_DME0 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 0 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CDMEIF0 LL_DMA_ClearFlag_DME0

LL_DMA_ClearFlag_DME1**Function name**

```
__STATIC_INLINE void LL_DMA_ClearFlag_DME1 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 1 direct mode error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CDMEIF1 LL_DMA_ClearFlag_DME1

LL_DMA_ClearFlag_DME2**Function name**

```
__STATIC_INLINE void LL_DMA_ClearFlag_DME2 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 2 direct mode error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CDMEIF2 LL_DMA_ClearFlag_DME2

LL_DMA_ClearFlag_DME3**Function name**

```
__STATIC_INLINE void LL_DMA_ClearFlag_DME3 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 3 direct mode error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CDMEIF3 LL_DMA_ClearFlag_DME3

LL_DMA_ClearFlag_DME4**Function name**

```
__STATIC_INLINE void LL_DMA_ClearFlag_DME4 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 4 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CDMEIF4 LL_DMA_ClearFlag_DME4

LL_DMA_ClearFlag_DME5

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_DME5 (DMA_TypeDef * DMAx)

Function description

Clear Stream 5 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CDMEIF5 LL_DMA_ClearFlag_DME5

LL_DMA_ClearFlag_DME6

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_DME6 (DMA_TypeDef * DMAx)

Function description

Clear Stream 6 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CDMEIF6 LL_DMA_ClearFlag_DME6

LL_DMA_ClearFlag_DME7

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_DME7 (DMA_TypeDef * DMAx)

Function description

Clear Stream 7 direct mode error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CDMEIF7 LL_DMA_ClearFlag_DME7

LL_DMA_ClearFlag_FE0

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_FE0 (DMA_TypeDef * DMAx)

Function description

Clear Stream 0 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CFEIF0 LL_DMA_ClearFlag_FE0

LL_DMA_ClearFlag_FE1

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_FE1 (DMA_TypeDef * DMAx)

Function description

Clear Stream 1 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CFEIF1 LL_DMA_ClearFlag_FE1

LL_DMA_ClearFlag_FE2

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_FE2 (DMA_TypeDef * DMAx)

Function description

Clear Stream 2 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CFEIF2 LL_DMA_ClearFlag_FE2

LL_DMA_ClearFlag_FE3

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_FE3 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 3 FIFO error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- LIFCR CFEIF3 LL_DMA_ClearFlag_FE3

LL_DMA_ClearFlag_FE4

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_FE4 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 4 FIFO error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CFEIF4 LL_DMA_ClearFlag_FE4

LL_DMA_ClearFlag_FE5

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_FE5 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 5 FIFO error flag.

Parameters

- **DMAx**: DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CFEIF5 LL_DMA_ClearFlag_FE5

LL_DMA_ClearFlag_FE6

Function name

```
__STATIC_INLINE void LL_DMA_ClearFlag_FE6 (DMA_TypeDef * DMAx)
```

Function description

Clear Stream 6 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CFEIF6 LL_DMA_ClearFlag_FE6

LL_DMA_ClearFlag_FE7

Function name

__STATIC_INLINE void LL_DMA_ClearFlag_FE7 (DMA_TypeDef * DMAx)

Function description

Clear Stream 7 FIFO error flag.

Parameters

- **DMAx:** DMAx Instance

Return values

- **None:**

Reference Manual to LL API cross reference:

- HIFCR CFEIF7 LL_DMA_ClearFlag_FE7

LL_DMA_EnableIT_HT

Function name

__STATIC_INLINE void LL_DMA_EnableIT_HT (DMA_TypeDef * DMAx, uint32_t Stream)

Function description

Enable Half transfer interrupt.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR HTIE LL_DMA_EnableIT_HT

LL_DMA_EnableIT_TE

Function name

```
__STATIC_INLINE void LL_DMA_EnableIT_TE (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Enable Transfer error interrupt.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR TEIE LL_DMA_EnableIT_TE

LL_DMA_EnableIT_TC

Function name

```
__STATIC_INLINE void LL_DMA_EnableIT_TC (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Enable Transfer complete interrupt.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR TCIE LL_DMA_EnableIT_TC

LL_DMA_EnableIT_DME

Function name

```
__STATIC_INLINE void LL_DMA_EnableIT_DME (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Enable Direct mode error interrupt.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR DMEIE LL_DMA_EnableIT_DME

LL_DMA_EnableIT_FE

Function name

```
__STATIC_INLINE void LL_DMA_EnableIT_FE (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Enable FIFO error interrupt.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- FCR FEIE LL_DMA_EnableIT_FE

LL_DMA_DisableIT_HT

Function name

```
__STATIC_INLINE void LL_DMA_DisableIT_HT (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Disable Half transfer interrupt.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR HTIE LL_DMA_DisableIT_HT

LL_DMA_DisableIT_TE

Function name

```
__STATIC_INLINE void LL_DMA_DisableIT_TE (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Disable Transfer error interrupt.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR TEIE LL_DMA_DisableIT_TE

LL_DMA_DisableIT_TC

Function name

```
__STATIC_INLINE void LL_DMA_DisableIT_TC (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Disable Transfer complete interrupt.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR TCIE LL_DMA_DisableIT_TC

LL_DMA_DisableIT_DME

Function name

```
__STATIC_INLINE void LL_DMA_DisableIT_DME (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Disable Direct mode error interrupt.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- CR DMEIE LL_DMA_DisableIT_DME

LL_DMA_DisableIT_FE

Function name

```
__STATIC_INLINE void LL_DMA_DisableIT_FE (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Disable FIFO error interrupt.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **None:**

Reference Manual to LL API cross reference:

- FCR FEIE LL_DMA_DisableIT_FE

LL_DMA_IsEnabledIT_HT

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsEnabledIT_HT (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Check if Half transfer interrupt is enabled.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- CR HTIE LL_DMA_IsEnabledIT_HT

LL_DMA_IsEnabledIT_TE

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsEnabledIT_TE (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Check if Transfer error nterrup is enabled.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- CR TEIE LL_DMA_IsEnabledIT_TE

LL_DMA_IsEnabledIT_TC

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsEnabledIT_TC (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Check if Transfer complete interrupt is enabled.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- CR TCIE LL_DMA_IsEnabledIT_TC

LL_DMA_IsEnabledIT_DME

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsEnabledIT_DME (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Check if Direct mode error interrupt is enabled.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- CR DMEIE LL_DMA_IsEnabledIT_DME

LL_DMA_IsEnabledIT_FE

Function name

```
__STATIC_INLINE uint32_t LL_DMA_IsEnabledIT_FE (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

Check if FIFO error interrupt is enabled.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- FCR FEIE LL_DMA_IsEnabledIT_FE

LL_DMA_Init

Function name

```
uint32_t LL_DMA_Init (DMA_TypeDef * DMAx, uint32_t Stream, LL_DMA_InitTypeDef * DMA_InitStruct)
```

Function description

Initialize the DMA registers according to the specified parameters in DMA_InitStruct.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
- **DMA_InitStruct:** pointer to a LL_DMA_InitTypeDef structure.

Return values

- **An:** ErrorStatus enumeration value:
 - SUCCESS: DMA registers are initialized
 - ERROR: Not applicable

Notes

- To convert DMAx_Streamy Instance to DMAx Instance and Streamy, use helper macros :
__LL_DMA_GET_INSTANCE __LL_DMA_GET_STREAM

LL_DMA_DeInit

Function name

```
uint32_t LL_DMA_DeInit (DMA_TypeDef * DMAx, uint32_t Stream)
```

Function description

De-initialize the DMA registers to their default reset values.

Parameters

- **DMAx:** DMAx Instance
- **Stream:** This parameter can be one of the following values:
 - LL_DMA_STREAM_0
 - LL_DMA_STREAM_1
 - LL_DMA_STREAM_2
 - LL_DMA_STREAM_3
 - LL_DMA_STREAM_4
 - LL_DMA_STREAM_5
 - LL_DMA_STREAM_6
 - LL_DMA_STREAM_7
 - LL_DMA_STREAM_ALL

Return values

- **An:** ErrorStatus enumeration value:
 - SUCCESS: DMA registers are de-initialized
 - ERROR: DMA registers are not de-initialized

LL_DMA_StructInit

Function name

void LL_DMA_StructInit (LL_DMA_InitTypeDef * DMA_InitStruct)

Function description

Set each LL_DMA_InitTypeDef field to default value.

Parameters

- **DMA_InitStruct:** Pointer to a LL_DMA_InitTypeDef structure.

Return values

- **None:**

79.3 DMA Firmware driver defines

The following section lists the various define and macros of the module.

79.3.1 DMA DMA CHANNEL

LL_DMA_CHANNEL_0

LL_DMA_CHANNEL_1

LL_DMA_CHANNEL_2

LL_DMA_CHANNEL_3

LL_DMA_CHANNEL_4

LL_DMA_CHANNEL_5

LL_DMA_CHANNEL_6

LL_DMA_CHANNEL_7

CURRENTTARGETMEM

LL_DMA_CURRENTTARGETMEM0

Set CurrentTarget Memory to Memory 0

LL_DMA_CURRENTTARGETMEM1

Set CurrentTarget Memory to Memory 1

DIRECTION

LL_DMA_DIRECTION_PERIPH_TO_MEMORY

Peripheral to memory direction

LL_DMA_DIRECTION_MEMORY_TO_PERIPH

Memory to peripheral direction

LL_DMA_DIRECTION_MEMORY_TO_MEMORY

Memory to memory direction

DOUBLEBUFFER MODE**LL_DMA_DOUBLEBUFFER_MODE_DISABLE**

Disable double buffering mode

LL_DMA_DOUBLEBUFFER_MODE_ENABLE

Enable double buffering mode

FIFOSTATUS 0**LL_DMA_FIFOSTATUS_0_25** $0 < \text{fifo_level} < 1/4$ **LL_DMA_FIFOSTATUS_25_50** $1/4 < \text{fifo_level} < 1/2$ **LL_DMA_FIFOSTATUS_50_75** $1/2 < \text{fifo_level} < 3/4$ **LL_DMA_FIFOSTATUS_75_100** $3/4 < \text{fifo_level} < \text{full}$ **LL_DMA_FIFOSTATUS_EMPTY**

FIFO is empty

LL_DMA_FIFOSTATUS_FULL

FIFO is full

FIFOTHRESHOLD**LL_DMA_FIFOTHRESHOLD_1_4**

FIFO threshold 1 quart full configuration

LL_DMA_FIFOTHRESHOLD_1_2

FIFO threshold half full configuration

LL_DMA_FIFOTHRESHOLD_3_4

FIFO threshold 3 quarts full configuration

LL_DMA_FIFOTHRESHOLD_FULL

FIFO threshold full configuration

MBURST**LL_DMA_MBURST_SINGLE**

Memory burst single transfer configuration

LL_DMA_MBURST_INC4

Memory burst of 4 beats transfer configuration

LL_DMA_MBURST_INC8

Memory burst of 8 beats transfer configuration

LL_DMA_MBURST_INC16

Memory burst of 16 beats transfer configuration

MDATAALIGN**LL_DMA_MDATAALIGN_BYTE**

Memory data alignment : Byte

LL_DMA_MDATAALIGN_HALFWORD

Memory data alignment : HalfWord

LL_DMA_MDATAALIGN_WORD

Memory data alignment : Word

MEMORY**LL_DMA_MEMORY_NOINCREMENT**

Memory increment mode Disable

LL_DMA_MEMORY_INCREMENT

Memory increment mode Enable

MODE**LL_DMA_MODE_NORMAL**

Normal Mode

LL_DMA_MODE_CIRCULAR

Circular Mode

LL_DMA_MODE_PFCTRL

Peripheral flow control mode

OFFSETSIZE**LL_DMA_OFFSETSIZE_PSIZE**

Peripheral increment offset size is linked to the PSIZE

LL_DMA_OFFSETSIZE_FIXEDTO4

Peripheral increment offset size is fixed to 4 (32-bit alignment)

PBURST**LL_DMA_PBURST_SINGLE**

Peripheral burst single transfer configuration

LL_DMA_PBURST_INC4

Peripheral burst of 4 beats transfer configuration

LL_DMA_PBURST_INC8

Peripheral burst of 8 beats transfer configuration

LL_DMA_PBURST_INC16

Peripheral burst of 16 beats transfer configuration

PDATAALIGN

LL_DMA_PDATAALIGN_BYTE

Peripheral data alignment : Byte

LL_DMA_PDATAALIGN_HALFWORD

Peripheral data alignment : HalfWord

LL_DMA_PDATAALIGN_WORD

Peripheral data alignment : Word

PERIPH**LL_DMA_PERIPH_NOINCREMENT**

Peripheral increment mode Disable

LL_DMA_PERIPH_INCREMENT

Peripheral increment mode Enable

PRIORITY**LL_DMA_PRIORITY_LOW**

Priority level : Low

LL_DMA_PRIORITY_MEDIUM

Priority level : Medium

LL_DMA_PRIORITY_HIGH

Priority level : High

LL_DMA_PRIORITY_VERYHIGH

Priority level : Very_High

STREAM**LL_DMA_STREAM_0****LL_DMA_STREAM_1****LL_DMA_STREAM_2****LL_DMA_STREAM_3****LL_DMA_STREAM_4****LL_DMA_STREAM_5****LL_DMA_STREAM_6****LL_DMA_STREAM_7****LL_DMA_STREAM_ALL****Convert DMAxStreamy**

__LL_DMA_GET_INSTANCE

Description:

- Convert DMAx_Streamy into DMAx.

Parameters:

- __STREAM_INSTANCE__: DMAx_Streamy

Return value:

- DMAx

__LL_DMA_GET_STREAM

Description:

- Convert DMAx_Streamy into LL_DMA_STREAM_y.

Parameters:

- __STREAM_INSTANCE__: DMAx_Streamy

Return value:

- LL_DMA_CHANNEL_y

__LL_DMA_GET_STREAM_INSTANCE

Description:

- Convert DMA Instance DMAx and LL_DMA_STREAM_y into DMAx_Streamy.

Parameters:

- __DMA_INSTANCE__: DMAx
- __STREAM__: LL_DMA_STREAM_y

Return value:

- DMAx_Streamy

Common Write and read registers macros

LL_DMA_WriteReg

Description:

- Write a value in DMA register.

Parameters:

- __INSTANCE__: DMA Instance
- __REG__: Register to be written
- __VALUE__: Value to be written in the register

Return value:

- None

LL_DMA_ReadReg

Description:

- Read a value in DMA register.

Parameters:

- __INSTANCE__: DMA Instance
- __REG__: Register to be read

Return value:

- Register: value

DMA_LL_FIFOMODE

LL_DMA_FIFOMODE_DISABLE

FIFO mode disable (direct mode is enabled)



LL_DMA_FIFOMODE_ENABLE

FIFO mode enable